EDA Report: Mutual Fund Analytics (Week 1)

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1. Objective and data

This report summarises the cleaning, validation and exploratory analysis of the 10 provided mutual fund datasets, and the key business insights found in them. The analysis was done in Python (pandas, NumPy, Matplotlib, Seaborn, SciPy). The full workflow is in `Python_Data_Analysis_Notebook.ipynb` and the cleaned files are in `data/processed/`.

Dataset	Rows	Columns	Content
fund_master	40	15	Scheme details: fund house, category, plan, expense ratio, risk
nav_history	46,000	3	Daily NAV for 40 schemes
aum_by_fund_house	90	5	AUM by fund house
monthly_sip_inflows	48	6	Industry SIP inflows and accounts
category_inflows	144	3	Net inflows by fund category
industry_folio_count	21	6	Industry folio counts (equity, debt, hybrid, others)
scheme_performance	40	19	Returns and risk metrics per scheme
investor_transactions	32,778	13	Investor-level SIP, lumpsum and redemption transactions
portfolio_holdings	322	8	Stock holdings and weights by scheme
benchmark_indices	8,050	3	Daily closing values of benchmark indices

2. Data cleaning

- **Whitespace:** no leading or trailing spaces were found in text columns.
- **Fund house names:** labels mixed "Mutual Fund" and "MF" (for example "SBI Mutual Fund" and "ICICI Prudential MF"). They were standardised to "Mutual Fund" (20 rows in fund_master, 45 in aum_by_fund_house, 20 in scheme_performance), which gives 10 consistent fund houses.
- **Dates:** stored as text in 9 of the 10 files (launch_date, date, month, transaction_date, portfolio_date). All were converted to datetime with 0 unparseable values.
- **Duplicates:** 0 duplicate rows in all 10 datasets.
- **Missing values:** only `yoy_growth_pct` in monthly_sip_inflows (12 of 48 rows, 25%). The first 12 months have no previous year to compare against, so this is expected and the values were left blank, not filled in.
- **Enrichment:** transactions were joined to fund_master on `amfi_code` to add scheme name, fund house, category, sub-category and plan. There were 0 unmatched transactions.

3. Data validation

All 8 validation checks passed.

Check	Result	Detail
NAV values are all positive	PASS	Minimum NAV 26.14
Transaction amounts are all positive	PASS	Minimum amount ₹400
fund_master amfi_code is unique	PASS	40 unique codes
Every fund_master code exists in nav_history	PASS	0 missing, and no extra codes in nav_history
Every transaction code exists in fund_master	PASS	0 unknown codes
Holding weights sum to about 100% per scheme	PASS	Range 99.98 to 100.02
No daily NAV move above 10%	PASS	Largest move 6.47%
No future dates in transactions	PASS	Latest transaction 30 May 2025

4. Outliers and anomalies

Outliers were detected with the 3 x IQR rule. They were flagged, not removed, because they may be genuine values.

- **SIP amounts:** 1,063 of 19,716 SIP transactions (5.4%) were flagged. The distribution is right-skewed (mean ₹11,018 against a median of ₹5,139, skewness 1.92, maximum ₹59,998), so these are large-ticket SIPs and were kept. No lumpsum or redemption amounts were flagged.
- **SIP new accounts:** April 2025 shows 46.0 lakh new SIP accounts, about 4 times the next highest month (10.5 lakh). The spike is unexplained and needs verification.
- **Category inflows:** the 12 highest values are all the Liquid category (about ₹32,000 to ₹42,000 crore a month). This is expected, as liquid funds receive far larger flows than other categories.
- **Max drawdown:** negative for all 40 schemes. This is expected because a drawdown measures a loss.
- **Sharpe ratio:** the highest value is ICICI Pru Liquid Fund - Regular (7.68). Liquid funds have very low volatility, which inflates the ratio, so Sharpe ratios should be compared within a category and not across categories.
- **Portfolio holdings:** 322 holdings cover 34 of the 40 schemes. The 6 schemes without holdings are all debt funds (2 gilt, 1 short duration, 3 liquid), which hold no stocks. Each scheme has only 9 to 10 stocks, so 149 holdings (46%) exceed 10% weight. All holdings share one snapshot date (31 Dec 2025), so this is a simplified sample, not full fund portfolios.

5. Key business insights

KPI	Value
Total transactions	32,778
Unique investors	5,000
Total investments (SIP + lumpsum)	₹227.71 crore
Total redemptions	₹124.45 crore
Net investment	₹103.25 crore
Total SIP amount	₹21.72 crore
Average SIP ticket	₹11,018
SIP share of transactions	60.2%
KYC verified	92.0%
Schemes with transactions	40

1. **SIP drives volume, lumpsum drives value.** SIP is 60.2% of transactions (19,716) but only 9.5% of the amount invested (₹21.72 crore). Lumpsum is 8,095 transactions worth ₹205.98 crore (90.5%), at an average ticket of about ₹2.54 lakh.
2. **Redemptions are large.** Redemptions of ₹124.45 crore equal 54.7% of investments. Net investment is still positive at ₹103.25 crore.
3. **Timing.** Net investment peaked in April 2024 and was lowest in May 2024.
4. **Geography.** T30 cities contribute 67% of SIP amount and B30 cities 33%. Madhya Pradesh is the top state by investment.

5. **Product mix.** Equity is the top category by investment.
6. **Age.** The 56+ age group has the highest average SIP ticket.
7. **Income does not explain SIP size.** Correlation between annual income and SIP amount is r = 0.003 (R^2 about 0.000, p = 0.673), so income is not a useful predictor of ticket size in this data.
8. **Payment modes are evenly split.** Net Banking (₹89.35 crore), Cheque (₹89.22 crore), UPI (₹88.82 crore) and Mandate (₹84.76 crore) each carry roughly a quarter of total transaction value, so there is no dependence on a single channel.
9. **Industry growth.** Monthly industry SIP inflow rose from ₹11,517 crore to ₹31,002 crore over the 48 months in the data (+169%).
10. **Performance.** The best 3-year return is SBI Small Cap Fund - Regular Plan - Growth (23.4%).

6. Statistical summary

SIP transaction amounts (₹):

Statistic	Value
Count	19,716
Mean	11,018
Median	5,139
Mode	579
Standard deviation	14,218
Skewness	1.92
Minimum	400
Maximum	59,998

The mean is more than twice the median, so most SIPs are small and a few are very large. The median is the better measure of a typical SIP. A simple linear regression of SIP amount on annual income gives a slope of ₹2.03 per extra lakh of income, which is not statistically significant.

7. Limitations

- The datasets cover different time ranges (for example transactions end on 30 May 2025 and holdings are a 31 Dec 2025 snapshot), so cross-dataset comparisons should be made carefully.
- Portfolio holdings are a simplified sample with 9 to 10 stocks per scheme.
- The results describe this provided sample and should not be generalised to the whole market.

8. Conclusion

The data is clean and consistent: no duplicates, all validation checks passed, and the only gaps and outliers have clear explanations. The analysis shows a business driven by lumpsum value and SIP volume, with large redemptions, a T30-heavy investor base and no link between income and SIP size. The cleaned datasets are ready for the SQL analysis and the Excel and Power BI dashboards.